

Design a Rate Limiter - Step 2: Propose High-Level Design & Get Buy-In

To keep things simple, we use a **basic client-server model** for communication.

1. Where to Place the Rate Limiter?

Rate limiter can be implemented in different locations:

Client-Side Implementation

- Implemented on the client itself
- **Not reliable** because:
 - Requests can be forged by malicious users
 - No control over client implementation

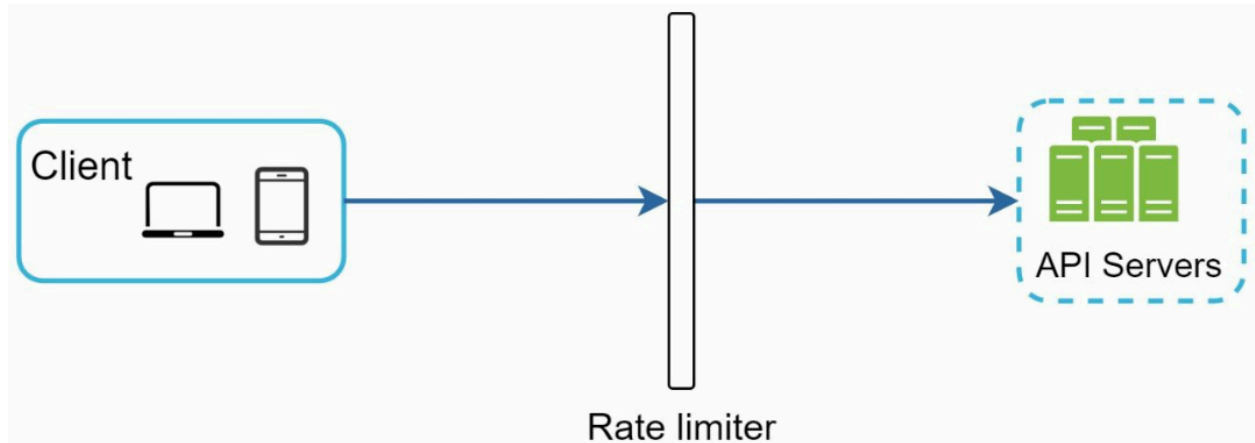
Server-Side Implementation

- Rate limiter is placed on the server
- Controls incoming requests before processing

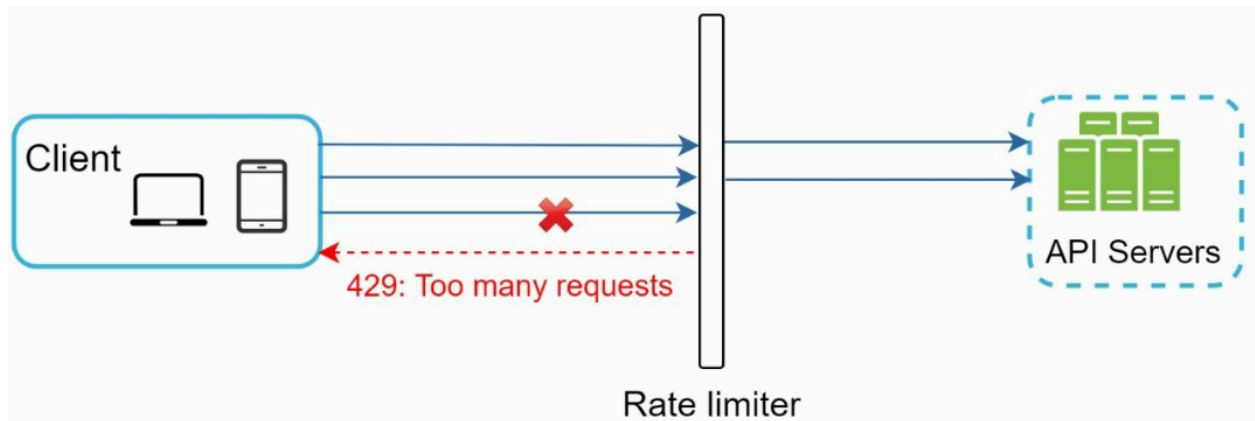


Middleware (API Layer)

- Rate limiter is implemented as a **middleware layer**
- Sits between client and API servers



2. How Rate Limiting Works (Example)



Assumption:

- API allows **2 requests per second**

Scenario:

- Client sends **3 requests in 1 second**

Flow:

- First 2 requests → forwarded to API servers
- 3rd request → blocked by rate limiter middleware

Response:

None

HTTP 429 → Too Many Requests

3. API Gateway Approach

- In modern systems (microservices), rate limiting is often implemented in an **API Gateway**
- API Gateway acts as a **middleware component**

Responsibilities of API Gateway:

- Rate limiting
- SSL termination
- Authentication
- IP whitelisting
- Serving static content

4. Design Decision: Server vs API Gateway

There is **no absolute answer**. It depends on system constraints.

Guidelines for Decision Making

✓ Technology Stack

- Check if your current stack (language, cache, etc.) supports efficient implementation

✓ Algorithm Flexibility

- Server-side → full control over algorithm
- Gateway → limited by provider capabilities

✓ Existing Architecture

- If using microservices + API Gateway → integrate rate limiter there

✓ Engineering Resources

- Building custom rate limiter → time-consuming
- Limited resources → use third-party API Gateway

5. Key Takeaway

Placement of rate limiter depends on:

- Control vs convenience
- Flexibility vs speed of implementation
- Existing system architecture